

Technology Stack

Date	7 July 2026
Team ID	SWETID-2026-5556
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Define Your Technology Stack:

To build a scalable, secure, and maintainable Credit Card Approval Prediction System, a robust modern technology stack is required. The chosen components handle seamless user registration, rapid data validation, instant integration with machine learning risk evaluation scripts, and real-time status notifications.

Technology Stack Details:

S No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	React.js, HTML5, CSS3	Chosen for building a clean, responsive applicant portal with high component reusability and smooth tracking interfaces.
2	Backend / Server-Side	Python / Flask (or Django)	Chosen because Python has native integration with Machine Learning frameworks, allowing fast API processing of credit risk models.
3	Database / Data Storage	MySQL (or PostgreSQL)	Provides strong relational data mapping to securely store structural applicant details, login credentials, and final evaluation choices.
4	Cloud / Hosting / Deployment	AWS (Amazon Web Services)	Ensures easy scalability to handle thousands of simultaneous applications, offering cost-effective and highly secure hosting.
5	Version Control & CI/CD	GitHub	Enables seamless collaboration across development branches, clear code tracking, and automatic deployment tests.
6	Third-Party APIs / Other Tools	Credit Bureau API & Twilio	Used to request instant credit scores from financial bureaus and send real-time SMS status notifications directly to applicants.